Dipanshu Saini

New Delhi | dipanshusaini271@gmail.com | 9643837403

LinkedIn | GitHub | LeetCode

EDUCATION

Year	Degree	Institute	CGPA/%
2024–2028	B.Tech (Mathematics and Computing)	Delhi Technological University (DTU), New Delhi	7.23 CGPA
2023	12th CBSE	Sarvodaya Bal Vidyalaya	82%
2021	10th CBSE	Akash Model Sr. Sec. School	90%

SUMMARY

B.Tech Mathematics and Computing student with hands-on experience in **Python, C++, AI-based application development, and machine learning**. Experienced in building practical applications, integrating LLM-based tools, data processing, and automation. Strong analytical and problem-solving mindset with an interest in software engineering, AI-first development, and building solutions for real-world problems.

PROJECTS

Teaching Buddy | Python, Streamlit, LangChain, Groq *GitHub / Live Link*

- Developed and deployed an AI-powered teaching assistant using Python and LLM-based tools for interactive learning and question answering.
- Implemented conversational memory to retain relevant context across user interactions and provide personalized assistance.
- Built an interactive Streamlit interface integrating AI capabilities into a practical end-user application.
- Integrated AI tools to automate repetitive educational tasks and simplify personalized learning workflows.

House Price Prediction | Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Built an end-to-end machine learning pipeline by cleaning and preprocessing housing data for model development.
- Performed exploratory data analysis to identify patterns and relationships influencing house prices.
- Applied feature engineering to transform raw data into relevant features for model training.
- Implemented and compared Linear Regression, Decision Tree, and Random Forest models for price prediction.

Email Classification | Python, Pandas, Scikit-learn

- Developed an email classification system using Python and Scikit-learn to automatically categorize text-based emails.
- Implemented text preprocessing including tokenization and stop-word removal to prepare email data.
- Converted email text into numerical representations using TF-IDF vectorization for machine learning models.
- Trained and evaluated multiple classification models to compare their predictive performance.

TECHNICAL SKILLS

- **Programming:** C++, Python
- **Problem Solving:** Data Structures & Algorithms, Problem Solving
- **AI & Development:** LLMs, LangChain, Streamlit, AI-assisted Automation
- **Machine Learning:** Scikit-learn, Regression, Classification, Model Evaluation
- **Data & Scientific Computing:** Pandas, NumPy, Matplotlib, EDA, Data Cleaning & Preprocessing
- **Tools:** Git, GitHub, Jupyter Notebook, Excel, Google Sheets